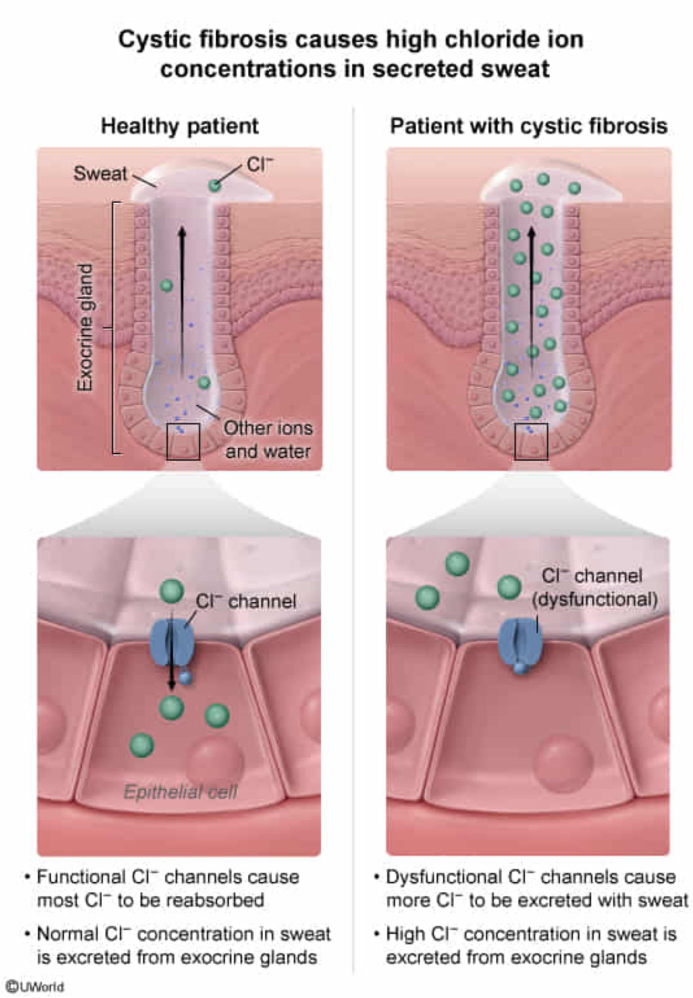
When present in exocrine gland cells, specific chloride channels reabsorb chloride ions from fluids in the exocrine gland lumen. Individuals with cystic fibrosis, an inherited disorder involving mucus buildup in certain tissues, exhibit decreased function of these chloride channels. An individual with cystic fibrosis in a hot environment is likely to demonstrate:

- ◯ A. high plasma chloride ion levels. [11%]
- ◯ B. low osmolarity of fluid in sweat gland lumens. [13%]
- ✔ ◉ C. high chloride ion concentrations in secreted sweat. [68%]
- ◯ D. low numbers of chloride channels in exocrine glands. [5%]

Correct
68% answered correctly

Explanation:



Exocrine glands (eg, sweat/sudoriferous glands) are glands in which the product (eg, sweat) is secreted through a duct onto the surface of the body (ie, skin) or onto the epithelial lining of a cavity inside the body. Most exocrine glands are composed of cells lining the open space (lumen) of the duct, which provides a pathway for the product to travel to the body surface or cavity. Membrane protein channels located in exocrine cells allow certain substances (eg, chloride ions) to travel between these cells and the lumen.

During thermoregulation in a hot environment, the body produces sweat, a hypotonic solution that is released onto the surface of the skin by sweat glands located in the dermis. When water from sweat evaporates from the skin surface, the body loses heat and is cooled in an endothermic reaction.

In this example, individuals with cystic fibrosis exhibit decreased function of specific chloride channels in the cells of sweat glands. The question states that these channels *reabsorb* chloride ions from fluids in the exocrine (sweat) gland lumen. If an individual with cystic fibrosis is in a hot environment, the body will thermoregulate by producing sweat.

Accordingly, this individual's sweat will contain a *high* concentration of chloride ions because *few* chloride ions will be reabsorbed by the dysfunctional chloride channels in sweat gland cells. Therefore, in a hot environment, an individual with cystic fibrosis will demonstrate **high chloride ion concentrations** in **secreted sweat**.

(Choice A) Because individuals with cystic fibrosis display decreased chloride channel function, many chloride ions will be *released* in secreted sweat. Therefore, the level of chloride ions that *remain* in the plasma will be low, *not* high.

(Choice B) Osmolarity refers to the solute concentration of a particular solution (eg, sweat). The fluid in sweat gland lumens (ie, sweat) of individuals with cystic fibrosis contains many chloride ions (solutes) that were not reabsorbed. Therefore, the osmolarity of the sweat will be high, *not* low.

(Choice D) The only information given is that chloride channels are *dysfunctional* in individuals with cystic fibrosis. There is *no* information regarding the quantity of these channels.

Educational objective:

Exocrine glands (eg, sweat glands) are glands in which the product (eg, sweat) is secreted through a duct onto the surface of the body (ie, skin) or onto the epithelial lining of a cavity inside the body. The skin accomplishes its function of thermoregulation through the production of sweat.

Time spent:0 sec	**QID**:403277
Subject:	**System:**
Biology	Skin and Immune Systems